

What each variable means — browsing features

Computed per participant from the Wikipedia free-browsing task
21 behavioural features · source: the browsing task (not the questionnaire)

Legend A. Browsing features and how each is derived

All features below come from each participant's own browsing trace.

Counts & searches

n_pages	Total article pages the participant opened while browsing.
n_unique_pages	Distinct articles opened (repeat opens counted once).
n_searches	Number of search queries the participant typed.
revisit_rate	Share of opens that were repeats: $(n_pages - n_unique_pages) / n_pages$.
search_vs_link_ratio	Searches per page opened; high = navigates by searching, low = by links.

Dwell — time on page

mean_dwell	Mean time per page in ms, derived from gaps between consecutive opens.
var_dwell	Variance of those per-page dwell times.

Semantic movement — from page-text embeddings

mean_step_distance	Mean cosine distance between consecutive pages' text; bigger = larger topical jumps.
var_step_distance	Variance of the step distances (how uneven the jumps are).
forward_flow	Mean semantic distance from each page back to all earlier pages (Gray 2019); higher = steadily diverging.

Session dynamics — within-session trend

dyn_slope	Trend of step distance over the session; negative = jumps shrink (explore→exploit), positive = grow.
dyn_early_late_delta	Mean step distance in the second half minus the first half.
dyn_n_steps	Number of semantic steps the participant contributed.

Browsing graph — visited-page network (Zhou 2024)

n_unique	Nodes in the visited-page graph (unique articles with cached text).
n_edges	Edges: a link from one visited page to another visited page.
clustering	Average clustering coefficient — how interlinked the neighbours are.
char_path_length	Mean shortest-path length within the largest connected component.
global_efficiency	Average inverse shortest-path — how efficiently the network connects.
lcc_fraction	Fraction of visited pages in the largest connected component.
path_breadth	Number of distinct Wikipedia categories the visited pages span.
bh_score	Busybody-Hunter composite (within-cohort z of edges+clustering+efficiency – path length); higher = more hunter-like.

What each variable means — creativity & cognition

External measures from the test battery, scored outside the browsing task

13 measures · source: the creativity/cognition battery (questionnaire & tests)

Legend B. Creativity and cognition measures and where each comes from

These are the person-level traits correlated against the browsing features.

Divergent thinking — fluency (number of answers)

AUT Broom - Number of Answers	Alternative Uses Task: count of unusual uses generated for a broom.
AUT Belt - Number of Answers	Alternative Uses Task, object = belt: count of answers.
AQT Pencil - Number of Answers	Alternative-uses-style task, object = pencil: count of answers.
AQT Pillow - Number of Answers	Alternative-uses-style task, object = pillow: count of answers.
Verbal Fluency - Number of Answers	Count of words produced in the verbal fluency task.

Divergent thinking — originality / quality

AUT Belt - Originality	Rated originality of the belt-use answers.
AUT Broom - Originality	Rated originality of the broom-use answers.
AQT Pencil - Originality	Rated originality of the pencil answers.
AQT Pillow - Originality	Rated originality of the pillow answers.
AQT Complexity Score	Complexity / elaboration score of the AQT answers.

Semantic forward flow — verbal

Verbal Fluency - Forward Flow	Forward flow on the generated words (Gray 2019): same construct as browsing forward_flow, in word space.
-------------------------------	--

Curiosity — self-report

Curiosity - Score	Self-reported curiosity questionnaire score.
-------------------	--

Cognitive control

GF - Score	Fluid-intelligence (Gf) score; included as a control for general cognitive ability.
------------	---

How they group in the analysis

Fluency and originality form divergent-thinking composites; verbal forward flow, curiosity and Gf enter on their own.

Composites follow creativity_model.py.

Per-participant browsing metrics: descriptive statistics

Every behavioural feature computed per participant, with its available N

N = 107 participants · 21 features · 9 features lose participants to inclusion criteria

Headline

All 21 metrics are shown for every participant (N = 107).

Counts and graph features keep all 107; semantic features need ≥ 2 vectors (N=100) and session dynamics need ≥ 3 steps (N=81). A red, bold N marks a reduced sample.

Table 1. Descriptive statistics for every per-participant metric

Mean, SD and range across participants; the N column exposes the inclusion criteria.

Feature	N	Mean	SD	Min	Median	Max	Inclusion criterion
n_pages	107	6.65	3.84	0.00	6.00	19.00	—
n_unique_pages	107	5.42	2.90	0.00	5.00	14.00	—
n_searches	107	5.60	3.49	0.00	5.00	20.00	—
revisit_rate	105	0.14	0.17	0.00	0.08	0.64	n_pages > 0
search_vs_link_ratio	105	1.00	0.61	0.11	1.00	4.00	n_pages > 0
mean_dwell	105	1.9e5	1.6e5	5.5e4	1.4e5	8.7e5	valid dwell
var_dwell	105	3.2e10	3.6e10	0.00	1.9e10	1.8e11	valid dwell
mean_step_distance	100	0.04	0.03	0.00	0.04	0.23	≥ 2 semantic vectors
var_step_distance	100	7.7e-4	0.00	0.00	3.3e-4	0.01	≥ 2 semantic vectors
forward_flow	100	0.05	0.03	0.00	0.04	0.17	≥ 2 semantic vectors
dyn_slope	81	1.5e-4	0.04	-0.12	3.2e-4	0.21	≥ 3 semantic steps
dyn_early_late_delta	81	-7.6e-4	0.03	-0.08	0.00	0.09	≥ 3 semantic steps
dyn_n_steps	107	5.32	3.73	0.00	5.00	18.00	—
n_unique	107	5.15	2.88	0.00	5.00	14.00	—
n_edges	107	4.30	4.38	0.00	3.00	18.00	—
clustering	107	0.22	0.30	0.00	0.00	1.00	—
char_path_length	107	1.15	0.66	0.00	1.20	3.05	—
global_efficiency	107	0.44	0.33	0.00	0.42	1.00	—
lcc_fraction	107	0.75	0.27	0.00	0.82	1.00	—
path_breadth	107	28.58	26.18	0.00	20.00	166.00	—
bh_score	107	1.4e-16	1.98	-2.75	-0.72	5.85	—

How to read

Each row is one browsing feature computed once per participant.

N = participants with a valid value; a red bold N means the criterion dropped some.

Full per-participant values are in participant_features.csv.

Browsing-creativity correlations significant before correction

Spearman ρ , $p < .05$ uncorrected, sorted by p (smallest on top)

$N = 100$ – 107 per test · 195 tests total · 0 survive BH-FDR

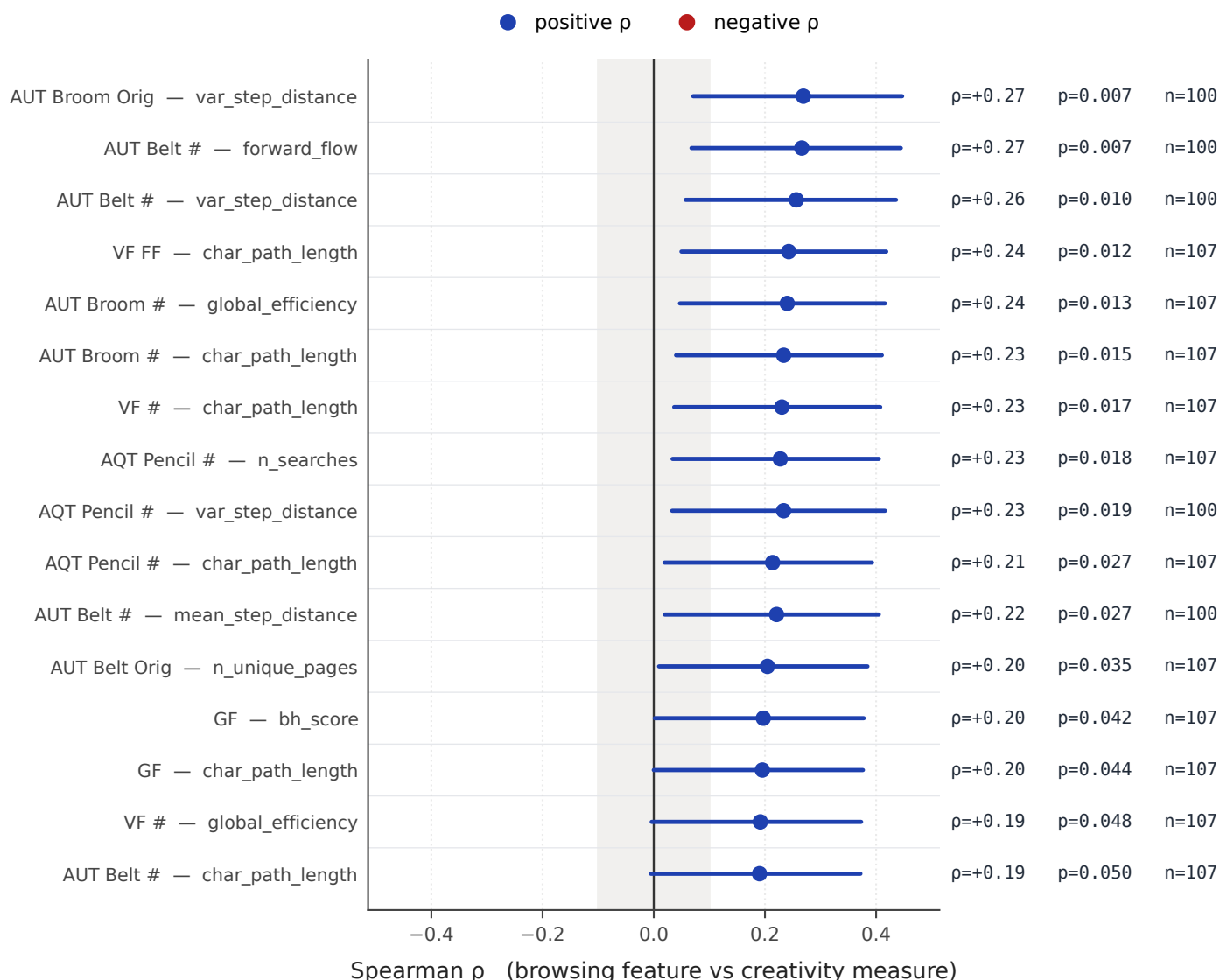
Headline

16 of 195 raw tests reach $p < .05$; chance alone expects ~ 9.8 .

All rows below are uncorrected. After BH-FDR, Bonferroni, or per-family FDR, none remain significant.

Figure 1. Effect size of each pre-correction hit (Spearman ρ , 95% CI)

Positive ρ = more of the feature goes with a higher creativity score.



How to read

Dot = Spearman ρ . Bar = its 95% CI (Fisher z, Fieller SE).

Line at 0 = no association. Shaded band = a negligible effect ($|\rho| < 0.1$).

Blue = positive association; red = negative. All rows are $p < .05$ uncorrected.

None survive multiple-comparison correction, so treat these as exploratory leads.